

# Closing the Pre-Infusion Gap: A Quality Improvement Initiative to Enhance Laboratory Readiness for Outpatient IV Iron Referrals

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Laboratory readiness increased from 69.1% to 89.8%

## MOTIVATION

At Renown in Northern Nevada, some patients arrived for IV iron infusions without valid iron studies, requiring same-day testing and coordination among nursing staff, pharmacy, and referring clinicians. Lab orders embedded in treatment plans did not reliably ensure readiness before arrival, highlighting the need for earlier verification.

★  
Reno

Nearly 1 in 3 lacked valid iron studies before infusion  
Baseline: 64/207 | Jan–Apr 2026

## AIM

Increase required iron study completion to ≥85% within six months among internal IV iron referrals, supported by verification before scheduling.

## ESTIMATED CHAIR-TIME BURDEN

Baseline period: Jan–Apr 2026  
Assuming 75 additional chair-minutes per visit lacking valid iron studies

- 20 excess chair-hours/month
- 240 excess chair-hours/year (annualized)

*\*Based on an infusion pharmacy staff estimate of 75 extra minutes per affected visit. Annual projection assumes unchanged baseline volume and frequency of missing or invalid studies; delays were not directly measured.*

## METHODS

Single-center QI evaluation of 416 internal outpatient IV iron referral records, January 1–September 14, 2026. All records were included regardless of treatment status.

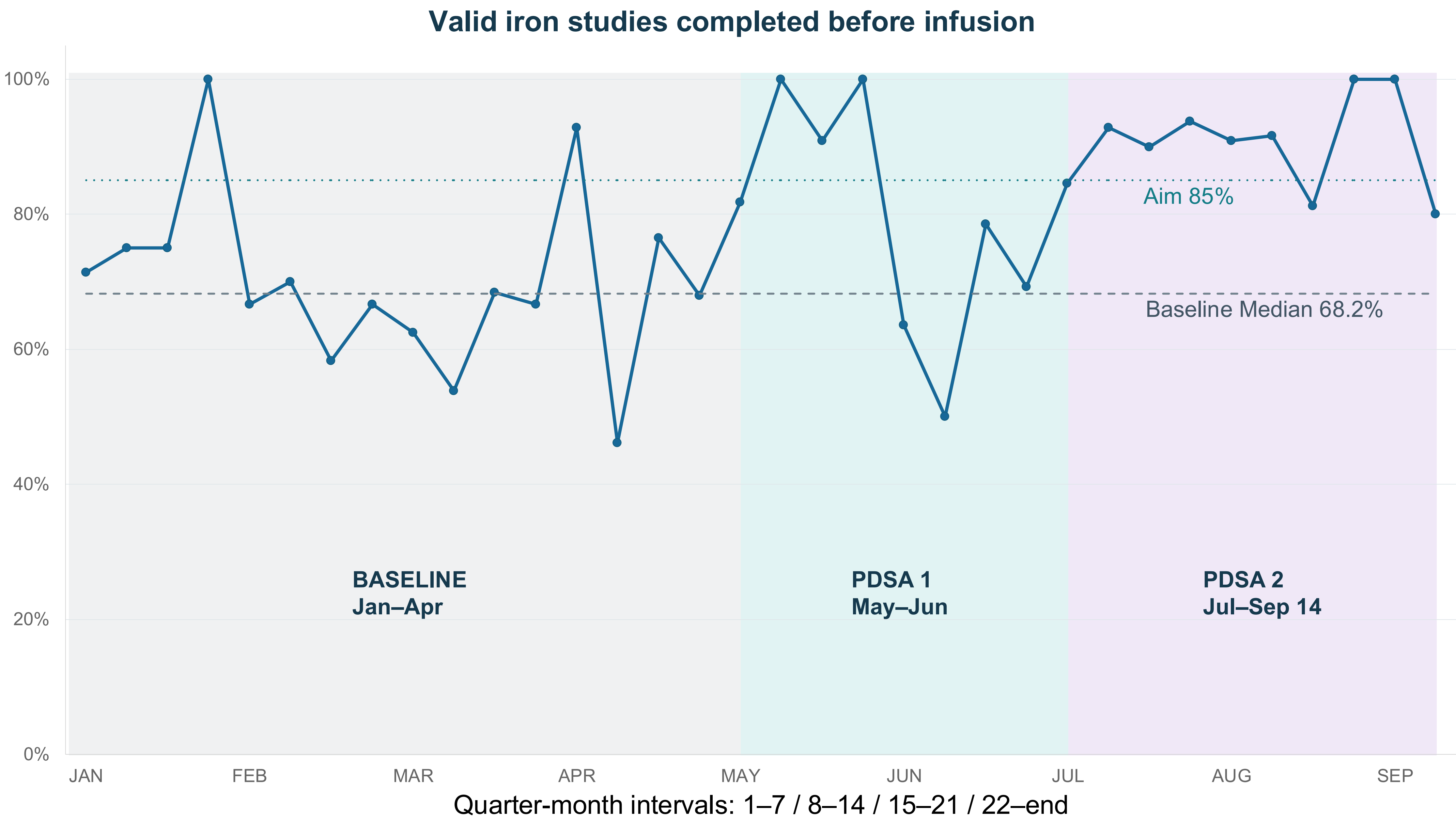
Process measure: Percentage with complete, valid iron studies before infusion.

Outcome measure: Percentage meeting laboratory treatment parameters.

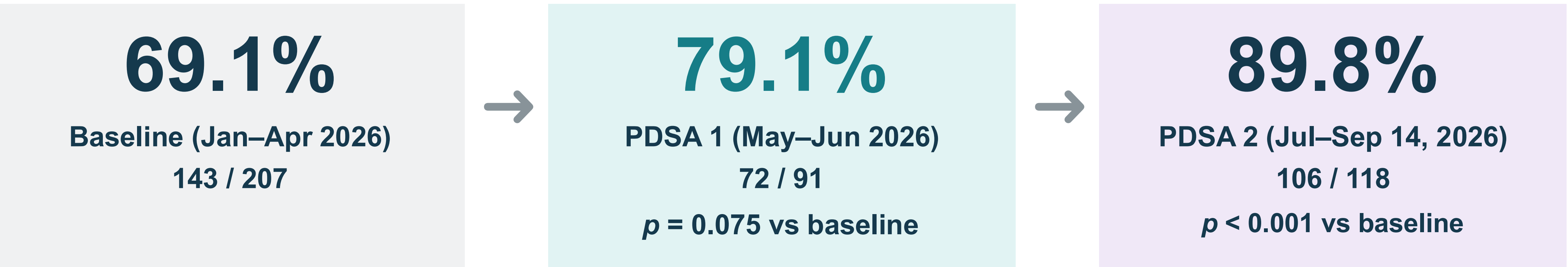
Analysis:

- Quarter-month run chart displayed process performance over time using a fixed baseline median.
- Pearson’s chi-square test compared pooled laboratory readiness rates across the three study periods: baseline, PDSA 1, and PDSA 2, assuming independent records. Outcome rates were summarized descriptively.

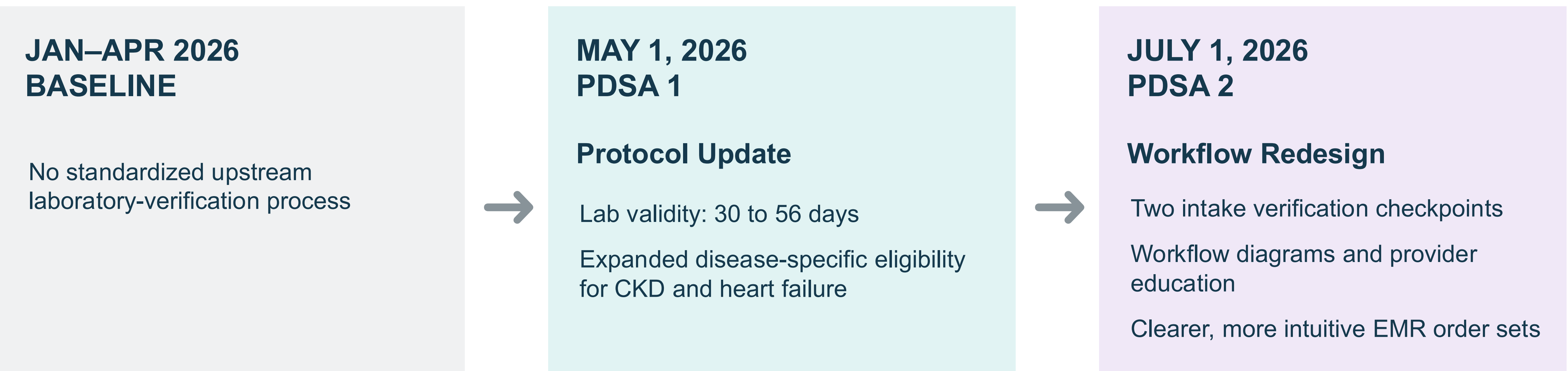
## PROCESS MEASURE: LABORATORY READINESS OVER TIME



## LABORATORY READINESS BY PERIOD



## WHAT WE CHANGED



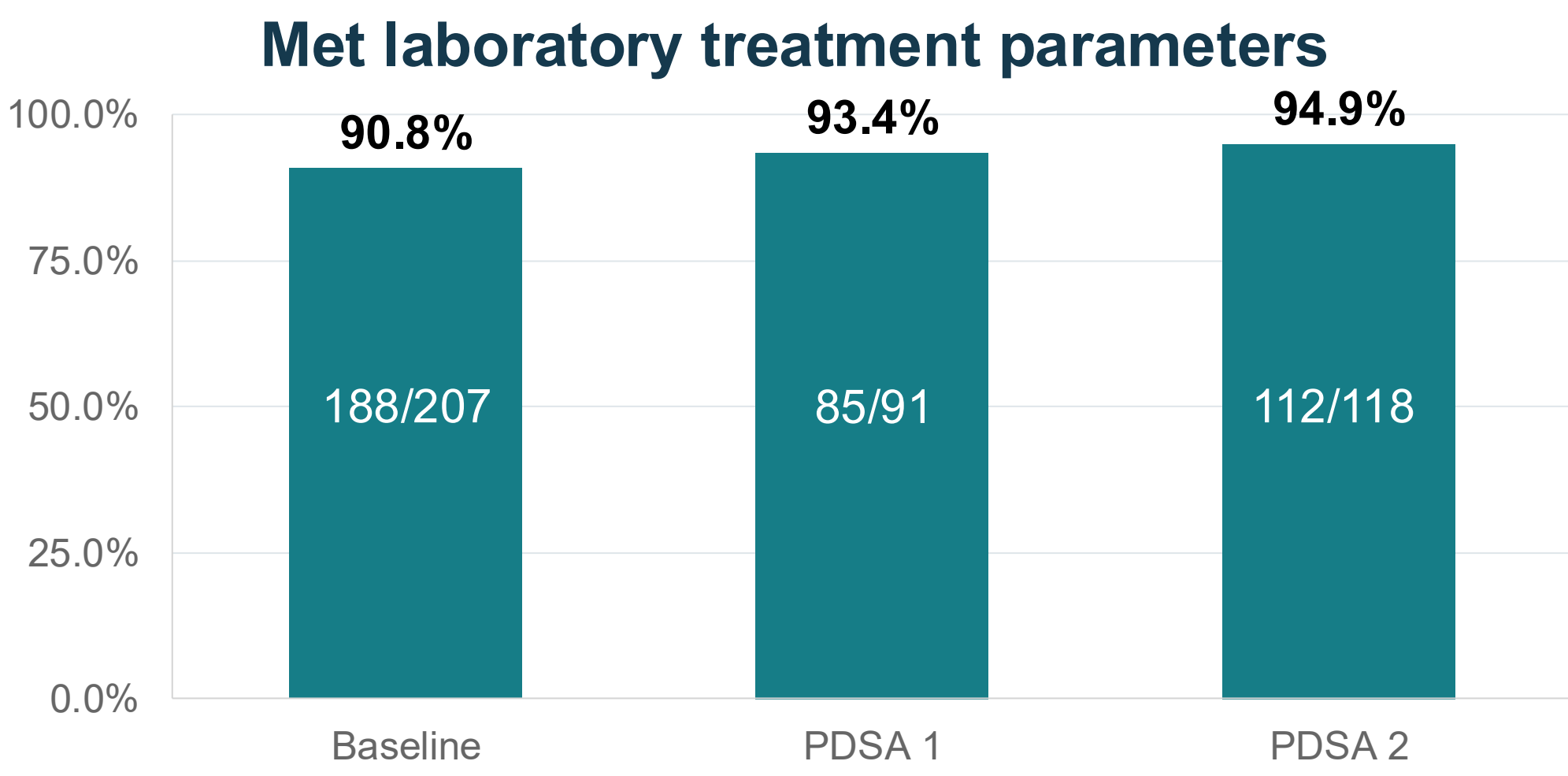
## ACKNOWLEDGMENTS

We thank Renown’s infusion nursing, pharmacy, and intake teams for implementation support, and American Society of Hematology mentors Drs. Ming Lim and Richard Godby for their guidance.

## REFERENCES

Renown Health. *Pricing Transparency*. Accessed September 30, 2026. <https://www.renown.org/Patients-and-Visitors/Billing/Pricing-Transparency>

## OUTCOME MEASURE



Attainment remained high across all periods.

## PATIENT IMPACT (ESTIMATED)

≈161 patient-hours of extra chair time potentially avoided annually\* from baseline to PDSA 2.

Potentially less waiting and fewer treatment disruptions for patients and caregivers, including those traveling from rural Northern Nevada.

## MODELED ANNUAL CAPACITY BURDEN

| Annual burden*                  | Baseline  | PDSA 1    | PDSA 2   |
|---------------------------------|-----------|-----------|----------|
| Excess chair-hours              | 240       | 162       | 79       |
| Illustrative opportunity value† |           |           |          |
| Medicaid                        | \$14,700  | \$9,900   | \$4,800  |
| Commercial mean                 | \$168,100 | \$113,500 | \$55,300 |
| Hometown Health National        | \$279,000 | \$188,000 | \$92,000 |

*Released chair time could expand access to oncology and other infusion therapies.*

*\* Assumes 621 annual visits, 75 extra chair-minutes per affected visit, and sustained period-specific rates; delay duration was not directly measured.*

*† Illustrative gross revenue proxy assuming one additional CPT 96365-eligible visit per released chair-hour and full utilization; not measured costs, lost revenue, or savings.*

## LIMITATIONS

Single-center, uncontrolled evaluation with limited follow-up. Changes in laboratory validity and eligibility criteria may affect comparisons across periods. Individual intervention effects cannot be isolated. Chair-time and financial estimates are modeled, not directly measured.

## CONCLUSION

Protocol and workflow changes were associated with improved laboratory readiness, exceeding the 85% target overall after PDSA 2. Laboratory treatment-parameter attainment remained high. Intermittent below-target intervals highlight the need for continued monitoring and refinement.

## NEXT STEPS

Monitor sustainability and adapt the workflow for an external-referral pilot.

